4 From automobility to sustainable transport

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Introduction

The problem for future urban transport emerges very clearly. It is a global dilemma, and unless the global perspective is maintained the dilemma will not be resolved. On the one hand, the global economy depends more than ever before on mobility. Intra- and inter-regional merchandise trade even after the global financial crisis stood at US$12.178 trillion in 2009 (WTO, 2010, Table 14): 12 trillion dollars worth of stuff moving around the world. There is, we can say, *embodied* mobility in almost everything we consume and produce today. On the other hand, that mobility depends overwhelmingly on fuels whose combustion is radically changing the biosphere through global warming. And it is time also to reconsider the more immediate and devastating personal costs of automobility to human health and life and to the health and life of local ecologies. A plague called traffic cannot be regarded as just a private matter. It is a public health issue. A recent article in the English language newspaper *China Daily* pointed out that 'in lower income countries the number one killer of people between 15 and 29 is road crashes' (Deng Fei, 2011). A recent epidemiological study of London and Delhi described the huge amounts of money that could be saved if sustainable transport came to replace automobility as the dominant paradigm (Woodcock et al., 2009).

The problem is global not only in the trivial sense that it occurs all over the world, but because the economic and ecological systems that ultimately make mobility possible are globally interlocking. For instance, it is no use imagining that electric cars will be a sustainable transport solution if, once taken up by the world economy – and especially by the emerging superpowers of China and India – it is found that there are not enough available mineral resources to build batteries for the world fleet. Nor is an electric – or hydrogen – solution sustainable if the power to generate the electricity or hydrogen is made by fossil fuels. So the first step in re-visioning urban transport must be to understand the fundamental problem as we have described it in the foregoing chapters of this book. The next step is to create an embracing vision of sustainable transport for the future that offers the hope of addressing the problem and resolving the dilemma. Before considering the dimensions of such a vision, however, we will review the vision we must leave behind, the vision which is still driving urban transport today along what we can now see is a road to perdition: automobility. It is a compelling vision, and it is worth considering not only what it stands for but the socio-technical process by which it was promoted and finally realized.

The cyclone of automobility

The vision of individual automobility is a phenomenon of the twentieth century. It offered freedom to move in unprecedented comfort, at unheard of speed, with an extraordinary level of personal control, an escape from the drudgery and danger of city life and the promise of reconnection with an idyll of nature. Moreover, after the mass production of cars was created by Ford and others, the vision was open to all to realize. The history of automobility is well documented in the USA, Europe and Australia, though different aspects of this history appear in different works. Sachs (1992), for instance, gives a particularly well-documented account of the rise of desire for independence, style and speed in Germany with less emphasis on the growth of the car industry. In the USA, Flink (1988) describes the rise of the automobile industry in great detail, but we look to McCarthy (2007) to tell us more about the consumption side: 'cars, consumers and the environment'. Davison with Yelland (2004) in Australia describe the political history of cars and roads. Dudley and Richardson (2000) examine road – and railway – policy in Britain from a political science perspective, and there are many other works touching on different aspects of the development of automobility (Wachs and Crawford, 1992; Sheller and Urry, 2000; Mosey, 2000; Vigar, 2002; Garrison and Levinson, 2006; Packer, 2008; Conley and McLaren, 2009). The historical record is therefore something like an incomplete jigsaw puzzle. The pieces for Asia and the new dynamic economies – for example, Brazil – are unfortunately missing, though there are some hints of significant local difference in the process, in Simon (1996), Dimitriou (1992, 2011), Karan and Stapleton (1997), Wu et al. (2007), and Imran (2010). But, however incomplete the picture, a pattern can be discerned which seems to apply at least in the industrialized world wherever automobility has evolved: a continuously self-reinforcing process in which public visions, private aspirations, technologies, industries, government regulation and infrastructure construction feed off one another, becoming powerful enough to overcome all countervailing forces to generate a structured path of human collective behaviour, something like a socio-technical cyclone.

How does such a cyclone start? No one can really know, but the conditions for its formation were certainly present for more than a century before the twentieth. Rifkin (2009: 335) tells us:

> As the tempo of industrialization picked up speed, urbanization accelerated and national governments consolidated their power, the 'rationalization' of government, market and social relations continued to keep

> pace. The period between 1790 and 1850 saw the increasing atomization of the individual in an ever more rationalized and integrated society.

Mobility and speed of travel were central to the integration of society composed of individual identities seeking pleasure and avoiding pain. The idea of speed was first realized with the steam locomotive, but with the invention of the automobile, speed could be appropriated individually by the rich. McCarthy (2007) begins his book on 'auto-mania' with a vivid description of the rapid adoption of the car for fun and sport by young American millionaires who 'owned and raced yachts, bred and ran thoroughbred, steeplechase, and harness racing horses, and played competitive polo, golf and tennis against one another. … Speed had nothing to do with getting somewhere sooner. Driving was simply fun'. Less fortunate Americans, he writes, could only 'gape in awe, envy and exasperation' (ibid.: 1). Likewise in Germany, as Sachs (1992: Ch 1) reports, automobiles provided 'pleasures for the wealthy'. Driving was an adventure, and the driver a sportsman 'taking up the reins of technology' (ibid.: 6). Interestingly the advertised imagery of cars, tapping the wellsprings of human desire, has changed little since the early days: geographical escape; social status; individual freedom; gender and sexuality – masculine: with cars sometimes portrayed as extensions of masculinity and sometimes as love objects; and power, speed and excitement (Conley, 2009: 39–40).

So the first cycle turned from the invention of a technology – automobiles, whether powered by steam, electricity or petrol (gasoline) – to its early adoption by the rich, and the creation of widespread envy and aspiration. As the reality and potential of cars became widely understood, there was strong resistance. A future President of the USA, Woodrow Wilson – then President of Princeton University – foresaw with deep consternation the outcome of automobility: 'I think, of all the menaces of today, the worst is the reckless driving in automobiles … In this the rights of people are set at nought'. Wilson asks how a person would react to the death of his child from being run over: 'I don't blame him if he gets a gun' (27 February 1906 cited by McCarthy, 2007: 12). McCarthy states:

> Wilson's remarks seem shocking, especially coming from a future U.S. president, but in 1906 they were typical of anti-automobile statements made by many elites, and compelling evidence not of a cavalier attitude on the part of Americans towards vigilante violence, but of the emotional intensity of the reaction to the automobile.
>
> Ibid.: 12

Similarly, in Germany, Sachs (1992: 13) reports the antagonism in the early twentieth century to the potential for cars to dominate the streets, especially of rural towns, which 'were after all inhabited by pedestrians, horse drawn conveyances of various types, children at play, and all kinds of fowl'.

In 1900 cars were banned from the streets of the rural Swiss canton of Graubünden.

Almost simultaneously, however, the aspirations of the many were turning people into consumers of automobiles with real interests just as the craft of automobile manufacture was becoming an industry with production interests. People wanted not just cars, for the reasons cited above, but also everything necessary for car travel: better roads, more available and cheaper fuel, parking space, accessories. As Flink (1988: 28) citing Interrante (1983: 90) observes, the consumption of cars 'satisfies a real need for transportation – a need as basic as food, clothing and shelter'. Aspirations of excitement gradually gave way to 'utilitarian advantages of economy and efficiency over the horse', which was, after all, the prime mover of the medieval transport revolution (Flink, 1988: 28). The British historian Tony Judt, who spent much of his working life in the USA, points out that the 'car culture' came to Europe in the 1950s, but, he writes, his own post-war generation 'grew up with cars and saw nothing distinctively appealing or romantic about them' (Judt, 2011: 42).

Car manufacturers wanted to make profits by selling cars. The petrol engine was first perfected in France and Germany giving French and German industries a brief advantage, but the principles of internal combustion were quickly taken up in Britain and the United States as cars and engines were imported. By 1904 American car production had overtaken that of France: 'France lost technological leadership as well to the United States with the appearance of moderately priced, light American runabouts in 1906–1908' (Flink, 1988: 25). Car manufacturers rapidly grew to be major employers. Ford wasn't the only manufacturer to see that higher wages would produce automobile consumers. The growth of consumption and production interests was accompanied by promotional activities such as car races, and, most importantly, the growth of a specialized press 'crusading' for mass adoption of the automobile (ibid.: 29). Flink noted the part played by the motoring clubs on both sides of the Atlantic in 'sponsoring tours and tests, lobbying for legislation favorable to motorists, and propagandizing the automobilists' cause' (p. 27) – and the 'close cooperation' between the press and the automobile industry (p. 30).

There is no need for conspiracy theories to explain the pressure put upon governments to support such a rapidly growing and massively employing industry, support mainly for building new and better roads for the cars to use: the 'good roads movement'. Flink reminds us that farming and rural interests in the USA were among those lobbying for better roads. At the Second Annual Good Roads Convention in 1909 at Cleveland, Ohio, the dominating influences 'were the American Automobile Association, representing the autoists and the cities, and the National Grange representing the farmers. Cooperating with these organizations were the American Road Makers' Association, the National Association of Automobile Manufacturers, and the American Motor Car Manufacturers' Association' (Flink, 1988: 170). But road building lagged demand. Not until 1930 was the primitive American road network, mostly unsurfaced and poorly sign-posted in 1904, transformed into 'an interconnected system of concrete highways' (ibid.).

Similar poor quality rural roads – in terms of the needs of the car – were normal all over Europe up to the end of the 1920s. Roads were, 'planned according to small scales, laid out for slow speeds, twisting along creeks and over hills, and emptying directly on to a market square, these streets were of use to bicyclists and horse carts but not fit for the space-mastering power of the automobile' (Sachs, 1992: 45). Sachs tells the story of the pressure of automobilism in Germany similar to that of America: 'the people's car and the freeway – the two pillars of a society on wheels' were to be realized in the 1930s, 'but not under the sign of democracy' (ibid: 46). The idea of a road just for automobiles linking cities across the country had been promoted by the construction industry in Germany in the 1920s, but Hitler's Reich led the way on motorway construction after 1933 with a view to stimulating the economy, creating jobs and unifying the nation.

After the War, America began the construction of a massive programme of interstate highways under the Interstate Highway Act of 1956, financed from the gasoline tax and other taxes on cars. Expenditure on roads between 1947 and 1970 was estimated to take up 75 per cent of all government expenditure on transportation (Flink, 1988: 176). Road building programmes were initiated in Britain, France and Italy as part of post-war reconstruction, and road building required expertise. Dudley and Richardson (2000: 86) note that the Minister of Transport, Boyd-Carpenter, 'sent delegations of engineers to Germany, the United States, France and Italy and so provided leadership to what became a key epistemic community, whose expertise would be required if the roads construction programme was to be accelerated'.[1]

In the 1920s only 2 or 3 per cent of people in the State of Victoria (Australia) held a driving licence. By 1950 most people in Australian cities still travelled around on foot, by bicycle or on public transport. It took a strong, sustained campaign by far-sighted people to convince governments that travel by car was the way of the future. Australian historians Davison and Yelland write about 'how the car won our hearts and conquered our cities' (Davison with Yelland, 2004). 'Motorists', they acknowledge, 'were a noisy lobby group, never reluctant to blow their own horns. Their opponents were a more diverse and defensive group, steeped in traditional attitudes and institutional loyalties, conscious of their declining patronage and often forced on to the back foot by the aggressive propaganda of motorists' (ibid.: 125). Australian town planners and road engineers visited the United States and returned with visions of freeways sweeping through the countryside. There was a strong aesthetic dimension: the sublime of engineering. Davison and Yelland (2004: 171) report the words of the Chairman of the Country Roads Board of Victoria: 'I really believe the freeway is a work of art … they are beautiful things, not evil things. They are things which enhance the beauty of the countryside not desecrate it'. But this was said in

1974, when building motorways through cities was under challenge from community groups who did indeed see the motorway as evil and the instrument of desecration.

By the time urban freeways were being built – and challenged – in the 1960s the problem of providing infrastructure for cars had changed radically. It was no longer one of bad roads but of traffic congestion on mostly very good roads. Good roads, cheaper cars and higher incomes enormously increased the volume of traffic, and with it the number of deaths and injuries from vehicle crashes. Cars, which were at first thought of as fun vehicles for speed and escape to the countryside, were replacing public transport for the journey to work in cities. New communities of experts grew up around the twin problems of road safety and the analysis of congestion.

When Woodrow Wilson warned of the future risks of a car-borne society he formulated the problem in terms of road safety and irresponsible driving which points to a solution: making roads safer to drive on, educating and regulating drivers, and making cars safer to drive *in*. This formulation does not challenge the idea of automobility itself, linked as it is with individual liberty. Road safety campaigns have been carried out in some form everywhere in the developed world alongside the growth of automobility. Mostly they have been very effective in reducing road trauma. In specifically US cultural terms, a 'Crusade for Traffic Safety' was launched in 1954, as Packer relates, 'by media men to help you, a media man, to save lives in traffic' (Packer, 2008: 28). It was led by President Eisenhower and involved a personal pledge to be signed by committed drivers, 'to work through my church, civic, business and labor groups to carry out the official action program for traffic safety' (ibid.: 29). However, Packer makes an important point:

> Reorganizing public opinion is an act of producing a new popular truth and in this case through a form of problematization. By this I mean that a phenomenon becomes recognized as a serious public problem through publicity that identifies it as such and suggests it be dealt with through expert explanation. In essence, it becomes a popular truth insofar as it creates a sense that reality is adequately represented, that something is amiss and must be set right, and, most generally, that there is a form of expertise that can sufficiently come to grips with the situation.
>
> Packer, 2008: 31

In a similar way, a popular truth was created about 'traffic congestion'. Traffic congestion is a sign of the very success of the 'private vehicles on public roads' solution to the desire for mobility. But congestion, a body metaphor, was portrayed as a real cost to the economy of a city, a region or a nation (see Low and Odgers, 2012). The corollary is that the cost of traffic congestion amounts to an economic return on public sector investment in infrastructure (see Box 4.1).

Box 4.1 Calculation of the cost of congestion

Saving travel time is a major part – usually the majority part – of the benefits calculated for infrastructure expenditure designed to reduce traffic congestion. The valuation of the cost to be saved is based on three assumptions: First, time spent in travel is time wasted – time that could otherwise be spent in productive activity. Second, if the 'wasted' travel time could be saved by investment in infrastructure, this economic value is equal to the monetary value of the additional income that the average road user would earn by reallocating travel time saved to income-earning activities. Third, a large part of the cost of congestion in a transport system is derived from the aggregate value of the many small time savings that could be made by all the individuals using the system if the infrastructure allowed something approaching 'free flow' of traffic at optimum speed.

This set of assumptions is problematic for five main reasons.

1 Time spent in travel is not necessarily wasted. Given a seat on a train or tram, many kinds of office work can be done: books, papers or reports can be read, telephone calls made, figures can be calculated, reports written on the laptop (Lyons et al., 2007).
2 The assumption that small portions of time saved by individuals can be aggregated to give a social value is highly questionable. Perhaps an hour extra at home or work, or even half an hour, has some experiential value to the individual, but does ten minutes extra have commensurate, or indeed any value? Might not the individual spend more time in bed or watching TV?
3 In order to calculate the cost of congestion it is necessary to have in mind a transport system that provides free flow against which to compare the existing situation. Congestion is a manifestation of the friction of distance: the obvious fact that it takes time to move from one place to another, but how much time is the minimum one can reasonably expect? How fast should one expect to travel? Very rarely, if ever, is the total cost of providing a transport system that can provide free flow under realistic assumptions of induced traffic entered into benefit-cost calculations involving congestion.
4 The assumption that any new road – i.e. any addition to existing road capacity – will directly result in higher average speeds is based on the proposition that since the new road adds to the road system's engineering capacity it enables an increase in average travel speed, and this increase in road capacity will not have a negative impact on vehicle flow, i.e. the number of vehicles that traverse a specific section of road per hour. Button (1993), however, notes that the speed–flow relationship is an unstable one and that, after the point at which a road's engineering capacity is reached, motorists 'often try and enter the road beyond this volume causing further drops in speed and resulting in the speed–flow relationship turning back on itself' (p. 111). Downs (2004: 102) concludes that, 'Once commuters realize the capacity of specific roads has been increased, they will quickly shift their routes, timing, and

modes of travel by moving to those roads during peak periods, thereby filling up the expanded capacity'. So, expanding the road network capacity, 'does not reduce the intensity of such congestion much in the long run' (ibid.).

5 There is now substantial evidence for the proposition advanced by Zahavi and Talvitie (1980) and Marchetti (1994) that people on average spend a more or less fixed – or constant – amount of time in daily travel – thought to be about an hour. If the speed of possible travel is increased, people travel further rather than reduce the time they spend travelling. Metz (2008: 6) cites data from the UK National Travel Survey to show that average per capita travel time in Britain remained more or less constant between 1970 and 2005, a period which saw enormous investment in new and improved roads. Over the same period the distance travelled increased.

Behind this formulation lies what Sachs (1992: 27) called an 'executive syllogism' of progress:

> What the critics of the automobile saw themselves confronted with in the debates of the time could be called the executive syllogism of competition-driven progress: a) technological development cannot be stopped; (b) escape is not an option, so Germany must take the lead; (c) therefore, we are called upon to support the automobile and its industry with all the means at the state's disposal. … In the face of so strong a consciousness of national responsibility, the critics became subdued and saw their questions – whether the automobile was necessary at all, and whether its advantages outweighed its disadvantages – grow strangely insignificant.

Sachs is writing about Germany at the beginning of the twentieth century, but the story of economic progress and, as traffic increased, that of the cost of congestion, together with an army of experts calculating that cost, lies at the heart of automobility everywhere. The way the problem was conceived was itself shaped by the technological solution: more and better roads for more and better vehicles. Storylines are tools for legitimating policy and coordinating action through coalitions comprising: 'the ensemble of a set of story-lines, the actors who utter those story-lines and the practices in which this discursive activity is based' (Hajer, 1995: 65). The history of the development of supportive transport 'storylines' has not been much recorded, but an Australian study, through documentary analysis and interviews, has identified the web of rationalization for road building in three Australian State capitals: Sydney, Melbourne and Perth over about 50 years (Curtis and Low, 2012).

To reiterate then, a vision of transport doesn't just appear unheralded on the political scene and immediately capture the public imagination and shape public policy. Nor is it a simple response to a technological opportunity. Rather, there is a process of generation in which a possibility triggers aspirations, tapping values, creating stories of explanation and problematization, shaping interests and epistemic communities, and then adding to technological opportunities – all of which support a system that grows and becomes more and more real so that it may sometimes appear to people in power, who are faced with the need to make decisions, that there is really no alternative (see Figure 4.1).

But of course there are alternatives. A cyclone can only exist while it draws on an environment that supplies it with energy from the heat and moisture of a sea. Once over land it loses its energy source. It continues under its own momentum for a time but it must gradually dissipate. So it is with automobility as a socio-technical structure. The source of energy – cheap fossil fuel – powering automobility, as we pointed out in Chapter 3, is coming to an end. But the dissipation of a cyclone does not mean the end of weather. Many of the same driving forces remain: the desire for mobility, the existence of an economy and society based around the mobility of goods and people, cities transformed by automobility, as we discussed in Chapter 2. If a vision of sustainable transport is to gather momentum and become accepted and influential, a socio-technical process of change will have to occur. That process will have some things in common with the process that generated automobility, but in some ways it will be different. Like automobility there

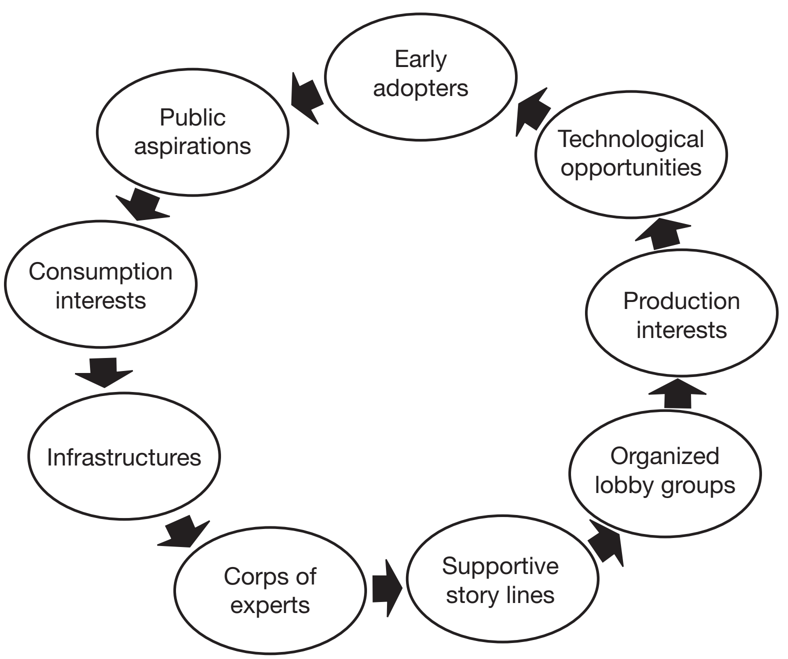

Figure 4.1 The cyclone of automobility (redrawn by Chandra Jayasuriya)

will have to be widespread support of a popular vision. A key difference is that the vision of sustainable transport is not so well aligned with the spirit of market capitalism as was the vision of automobility.

The new vision of sustainable transport

The origins of 'sustainable development'

The idea of sustainable development originated after it began to be understood some 50 years ago that pollution from industrial growth was doing serious damage to the natural environment, and this was not just a matter of local pollution but a global concern. The first images of Earth seen from space powerfully reinforced the idea of a vulnerable planetary environment. But the governments of developing countries meeting at the United Nations General Assembly became concerned that worries in the richer countries about global pollution would harm their own prospects of industrial growth and their chance to generate wealth. So the UN initiated a series of conferences on environment and development, the purpose being to set norms which would steer the world towards forms of industrial growth that would not threaten the quality and integrity of the global natural environment. Hence the Stockholm Conference of 1972 'considered the need for a common outlook and for common principles to inspire and guide the peoples of the world in the preservation and enhancement of the human environment' (UNEP, 1972). The subsequent UN declaration proclaimed that:

> In our time, man's capability to transform his surroundings, if used wisely, can bring to all peoples the benefits of development and the opportunity to enhance the quality of life. Wrongly or heedlessly applied, the same power can do incalculable harm to human beings and the human environment.
>
> Ibid.

The language is anthropocentric. The reports speak of the 'human environment' with 'Man' at the centre. But the underlying idea of 'sustainable development' that came out of these deliberations (including the World Commission on Environment and Development, the Brundtland Commission: WCED, 1987) was that humans can change the nature of industrial development, but they cannot change the limits imposed by the natural world.

However, as consensus was pursued on conflicting aims, the meanings given to 'sustainable development' proliferated. As Sachs (1999: 77) observed, 'Dozens of definitions are being passed around among experts and politicians, because many and diverse interests and visions hide behind the common key idea'. The price for consensus, Sachs observes, was therefore lack of clarity. This lack of clarity allowed 'sustainability' in the language of government to become a green cloak for business as usual. Sustainability, or 'sustainable development' has come to be conventionally defined in terms of a tripartite idea involving the economy, society and environment. While there are many different theoretical formulations of the tripartite idea, the dominant formulation, at least as it appears in public policy, encourages the belief that all three dimensions, economy, society and environment, *must all* be sustained. In lay terms, sustainability therefore now tends to mean that we have to keep the economy thriving, we must hold society together, and we must not deplete the environment too much. In such a formulation sustainable development becomes a political compromise between essentially conflicting ecological, social and economic goals (Low, 2003: 2–4).

As the current global financial crisis unfolds, however, the conflict between business growth, social integration – in Europe – and environmental conservation becomes patently obvious. The escalating price of oil is stimulating a worldwide search for alternative sources of fossil fuel. As Krauss (2011) puts it:

> From the high Arctic waters north of Norway to a shale field in Argentine Patagonia, from the oil sands of Western Canada to deep-water oil prospects off the shores of Angola, giant new oil and natural gas fields are being mined, steamed and drilled with new technologies.

No word about environmental limits there!

Over the 25 years since the report of the World Commission was published it has become much clearer that economic growth – that is growth of business activity – is negatively affecting the global environment. The emission of carbon gases to the atmosphere triggering global warming is only one symptom of the disturbance to natural ecological cycles caused by economic growth (Steffen et al., 2005). As Rifkin (2009: 478) starkly warns, 'Global warming is the entropy bill for the Industrial Revolution'. Despite 'sustainable development' having become a popular mantra in government circles almost everywhere, the world continues on a path of ever greater environmental destruction. We need to start talking about global entropy. Instead of the word 'sustainable' denoting what it was originally intended to mean: economic growth within environmental limits, it has commonly become a way of justifying growth with only minor environmental palliatives. Instead of the environment providing the limiting framework for economic growth, economic growth has become the non-negotiable condition for global environmental conservation.

While the environment could be considered a limitless resource, social and economic development could be thought of as separate matters from environmental conservation. The latter was seen as proceeding from and subject to society and politics, with, for instance, environmentalist groups and Green Parties representing a somewhat marginal social interest in Nature. Now, however, as natural resource limits are approached

– including the capacity to absorb waste – it becomes important to think about the economy as an 'entropic flow': a one way flow through a system of production beginning with resources extracted from the environment and ending in waste put back into the environment. Waste, by definition, is stuff that cannot be reused to create the same level of value as the original resource – CO_2 being a pure example. This is the concept developed by Nicholas Georgescu-Roegen (1971) and further explored by Daly (1996). This concept places the economy firmly within the environment and subject to its limits in a way that, surprisingly perhaps, conventional economics does not entertain.

Daly compares economics with biology. Conventional economic doctrine, he says, 'is exactly as if a biology textbook proposed to study an animal only in terms of its circulatory system, without ever mentioning its digestive tract!' Through the digestive tract, animals 'continuously take in low-entropy matter/energy and give back high entropy matter/energy ... But in economics there is only the circulatory system. This is as true of Marxist economics as it is of neoclassical economics'. He asks, in tutorial mode, 'What concept in economics ties the economy to its environment? Circulation of blood is to circulation of money as the digestive tract is to ... (*what*?)' (Daly, 1996: 193). Climate change resulting from human-generated CO_2 emissions is an irrefutable indicator that an environmental limit has not only been reached, but exceeded. It is not the only such indicator but certainly a very important one for an economy and a transport system heavily dependent on burning fossil fuel. To be meaningful, the term 'sustainability' has to take on board the understanding that the economy, and the society that generates and sustains it – fundamentally, liberal capitalism – is today subordinate to environmental limits.

It should be stated at this point that the above approach to sustainability is compatible with 'multi-criteria' analysis, which assesses policies against environmental, social and economic criteria. Such analysis is legitimate and appropriate just so long as the criteria are kept separated. An example is the work being done in Europe on traction batteries under the SUBAT research programme entitled 'Sustainable Batteries' (Van den Bossche et al, 2005). What is not legitimate, which to its credit the research appears to have avoided despite the misuse of the term 'sustainable', is to bundle all the criteria together and present the bundle as a measure of sustainability. Multi-criteria analysis has the valid aim of finding the most cost-effective, socially fair and technically feasible pathways to an economy that functions within environmental limits. The approach is compatible with Daly's formulation.

Daly's is essentially a prudential approach based on the idea that we have to find practical ways of allowing economic efficiency – so that supply meets demand – and socially fair distribution of wealth to occur within environmental limits. But the idea of sustainability might also be thought of as being imbued with moral purpose: in positive terms conserving the Earth's ecosystems and species evolved over billions of years, or in negative terms avoiding a mass species extinction event (Lynas, 2008: 250). Seen in this light, compromising with sustainability becomes suspect. Do we expect physicians to temper the principle of preserving human health by also supporting the tobacco and alcohol industries? Do we ask lawyers to dilute the principles of justice on the grounds that some interest groups will be hurt by the application of the principles? Do we condone engineers departing from the principle of structural equilibrium when designing a bridge or an engine in order to allow their clients to enhance their profits? The answer of course is 'No'. If unprincipled behaviour sometimes occurs, it is rightly regarded as reprehensible or even corrupt. Seen in this light, those who claim to speak for sustainability are subject to a similar sanction. In view of what we now know about the danger to the world's ecological systems of 'business as usual' economic growth it seems reasonable and natural to think of sustainability as a moral imperative like justice or human health, and for similar reasons. The question of environmental justice in the sense of justice to Nature – ecological justice – is a complex matter which has been explored in depth in the literature of philosophy (see Low and Gleeson, 1998, Chapter 6). Here is not the place to expand on this issue, but the basis of all such moral imperatives is the recognition of the intrinsic value of life: human life, of course, but in the case of environmental justice extending the value of life beyond the human species.

Sustainable transport principles

There are many different values that a transport system should serve other than pure mobility: for instance, public health and safety; economic efficiency; economic development; distributional fairness – or social justice; community integrity; and quality of place – as discussed below. Most are compatible with – and may be supportive of – the value 'sustainability'. But let us call them by their name instead of subsuming all under the one word 'sustainability'. The word 'sustainable' refers to sustaining the global environment.

It is much easier to identify *unsustainable* development than to define sustainability because all human exploitation of the environment necessarily impinges upon it and changes it. It is difficult to specify the limits to economic activity in terms of a broad ideal of sustainability, let alone turn those limits into operational indicators. But it is necessary to do so in order to restore some clarity to the term 'sustainability'. Fortunately both the limits and the indicators with respect to global warming are clear and scientifically established (Stern, 2006; Garnaut, 2008; Pachauri and Reisinger, 2007). So we use those limits as the basis of a simple and unambiguous statement of principle. The first principle of sustainable transport can therefore be stated thus:

- *For an urban transport system to be sustainable it must be one in which the carbon emissions from its operation, and embodied in new*

> *infrastructure construction, are reduced to a level compatible with a global temperature rise from pre-industrial levels of no more than 2 degrees Celsius.*

But this principle necessarily has global reach, if only because the atmosphere is a global system, a global commons, and every country's emissions affect the whole system. So we now have to consider a principle of justice as fairness (Rawls, 2001). In such a case it seems reasonable and simple to apply the 'categorical imperative' version of the 'golden rule' enunciated by Immanuel Kant, 'Act only according to that maxim whereby you can at the same time will that it should become a universal law without contradiction' (Kant, 2002 [1798]). The principle finds expression in Article 1 of the Universal Declaration of Human Rights: 'All human beings are born free and equal in dignity and rights' (United Nations, 1948). The Article embodies the concept of equal human rights more generally, including the equal right to property on which the global economy is supposed to be based. Thus the right to pollute the atmosphere in order to generate economic growth is one that applies equally to all people. Starting the application of that principle from the present is not perfectly fair because some nations discovered and followed the path to growth by burning fossil fuel earlier than others, in the process filling the atmosphere with long lived greenhouse gases. Developing nations justifiably grumble that it is the past industrial growth of the developed world that has already loaded the atmosphere with carbon gases. However, retrospective application of a law-like principle of justice comes up against another principle: that people have to know the consequences of their action. Plainly in the case of global warming, the consequences were not known until comparatively recently. These ideas were canvassed in some detail by Baer et al. (2000). Nevertheless, there is an obligation to move towards equality. If then we apply the Kantian imperative, we can state the second principle of sustainable transport thus:

- *For a sustainable urban transport system also to be fair it must be one whose per capita carbon emissions are approximately the same in all cities worldwide.*

These are basic minimum principles of sustainable transport. They are rigorous and will be difficult for rich nations, or rapidly developing nations, to attain. It is important to stress that actions to transform transport almost certainly cannot result rapidly in 'sustainability'. Sustainability is more like a compass point that gives a direction to policy rather than an absolute criterion. But it is more useful for policy to have a meaningful direction than a meaningless criterion. Eventually the direction must become a criterion. The question, of course, is how quickly must this convergence occur? Based on well-established climate science, all transport systems except those in the world's poorest cities are currently unsustainable. The aim for the rich cities of the world, and for the rich socio-economic sectors within them (as we discussed in Chapter 3), must be to move towards zero carbon transport systems within a period of time which would protect the world from dangerous climate change.

Sustainable transport practices

Much more has been written on the practices of sustainable transport – or sustainable mobility – than on the principles these practices are meant to serve. Two recent books on sustainable transport (Schiller et al., 2010; and Black, 2010) include as a condition for sustainability that emissions and waste must be limited to the Earth's capacity to absorb them. Schiller et al. (2010: 2) cite the University of Winnipeg's definition in which the above condition appears to rank equally with such desiderata as the human need for access – presumably to the places people need to go to from home in order to get what they want; human and ecosystem health; equity within and between generations; operational efficiency; and a 'vibrant' economy – vibrating not too much we hope. Black (2010: 4) turns, as we do, to Daly, again noting the sustainability condition: 'the rate of pollution emission does not exceed the assimilative capacity of the environment'. Black says that Daly (1992) does not define what he means by sustainability – though the conditions Black cites surely come close – but in 1996 he does so, specifying sustainability as an economic concept in which the economy is viewed as an open system (Daly, 1996: Ch 2). Daly's analysis is sound and it is this on which we rely to arrive at the basic principles stated above (see Box 4.2).

Having adopted Daly's sustainability condition, and in awareness of the risks of global warming and the IPCC report (2007), Black remains ambivalent about the merits of trying to reduce carbon emissions from transport. Concluding a section on 'Climate Change', he writes:

> Abatement was briefly mentioned, but dismissed as something that does not now seem like a possible strategy for limiting the impacts of climate change. Nevertheless, it does make some sense to try and limit the emissions of greenhouse gases to prevent the impacts from being worse than currently anticipated.
>
> Black, 2010: 35

Exactly what the difference is between 'abatement' and 'mitigation' is unclear. Later he argues that, 'we do not understand the feedback mechanisms at work in terms of the interaction of atmospheric carbon and oceans', or what the resulting increase in temperatures will be. He discusses the impacts of global warming and the attendant uncertainties and writes, 'Until we have a clearer picture of these impacts, incorporating the costs of transport's share of global warming emissions [in the list of costs of automobility] does not appear to have much merit' (ibid.: 92). The consequence of such a view is likely to be continued inaction until it is too late to act to abate global warming whose upper limit will only be reached once the

Box 4.2 Herman Daly's positioning of sustainability as an economic concept

Daly (1996: 48–52) describes the global economy as an open system with inputs – resources – from and outputs – waste – into the environment. He ranks 'sustainability' as a macroeconomic economic concept comparable with 'allocation' and 'distribution'.

Efficient allocation via markets ensures broadly that what people desire – always under the constraint of what they have available to spend – matches what businesses produce. Fair distribution means that there is justice in the distribution of that very capacity to spend. Of course what is social justice in distribution is a contested notion. But governments in democracies today have accepted the task of ensuring that nobody lives in absolute poverty, and that means accepting a concept of fair distribution – or at least limits to unfairness.

In Daly's view, 'sustainability' as a macroeconomic concept relates to the scale of the economy – production, consumption and distribution – in terms of what it takes from and puts into the planetary environment. The important assumption here is that the scale of what is taken from and put into the environment is limited by the environment's capacity to supply inputs and accept waste. Sustainability has come to be understood as an economic concept as we approach a 'full world' – one in which the economy is pressing upon its environmental limits. Global warming is an indicator that one such limit has been exceeded – the limit of a stable climate. Bluntly, nine billion people probably cannot survive on a substantially hotter planet

Daly (op. cit.: 51–52) further distinguishes between a maximum scale and an optimum scale. The optimum scale can be further divided into the 'anthropocentric optimum' and the 'biocentric' optimum. The former is, 'the point at which the marginal benefit to human beings of additional man-made physical capital is just equal to the marginal cost to human beings of sacrificed natural capital'. Natural capital includes environmental 'sinks' absorbing waste. The latter is where 'other species and their habitats are preserved beyond the point necessary to avoid ecological collapse or cumulative decline'. The biocentric optimum is based on recognition that other species have intrinsic value independent of their value to humans.

entire ecology of the planet is transformed. Beyond warming above about 3°C above pre-industrial times, the 'carbon cycle feedback' may begin (Cox et al., 2000). Lynas's explanation of the findings of Cox and his colleagues at the UK Hadley Centre is clear, and worth citing at length.

> Why should the paper, published in *Nature*, have rung alarm bells? First, it showed that global warming could begin to generate its own momentum if a previously unforeseen positive feedback – a vicious circle by which warming would release more greenhouse gas, causing more warming and thereby more gas to be released in an unstoppable

> spiral – came into effect. This, the 'carbon cycle feedback' referred to in the paper's title, would potentially leave human beings as powerless bystanders in a devastating runaway global-warming scenario.
>
> Lynas, 2008: 138

The feedback potential discussed by Cox et al. is not by any means the only paper on the subject. Of course there are uncertainties. That is why climate scientists talk about 'risk'. But at least some of these uncertainties are greatly magnified in the popular media by scientists and others in the service of vested interests in fossil fuel (Oreskes and Conway, 2010).[2] Enough is understood about climate change and published in the scientific literature to indicate that the risks are incalculably large and abatement is the number one issue.

Both Schiller et al. and Black, as well as a host of earlier texts (for instance Banister, 2005), move on fairly quickly, and indeed very usefully, to a variety of problems of *unsustainable* transport and the practices which should be adopted to deal with them. We can only briefly summarize here the quite complex discussions around the kinds of practices advocated under the rubric of sustainable transport.

Public transport

(see Images 4.1.1 to 4.1.19 in the Companion Website)
Most texts on sustainable transport support increased use of collective or public transport – or 'transit' in American terms – for travel both within and between cities (Whitelegg, 1997: 188; Newman and Kenworthy, 1999; OECD, 1995, 2002; ECMT, 2003; Schiller et al., 2010: 96; Humphreys, 2011: 117; and see Chapter 3 above). Glover (2011) posits that public transport is a common pool resource. All of these texts see a shift to public transport as just one among a range of multifaceted policies (33 in Whitelegg, 1997: 204–05). Most couple the necessary mode shift with increased restrictions on car use. Most also discuss the institutional changes required to make the shift possible. In the US context, Black (2010: 270) observes:

> There are some indications that attractive public transit options are proliferating around the United States. If we are ever going to manage the congestion problem, we must have this alternative mode available for use. Little attention has been given to public transit here [in his volume]. In large part this is due to the fact that a doubling of transit capacity would at best double transit ridership. In the United States this progression from 3% to 6% of all trips would not, in and of itself, solve the sustainability problem – but it would contribute to the solution.

We can agree strongly with the second claim – on transit and sustainability – while still questioning the first, for gaining ridership depends on much more than simply capacity, as Vuchic argues (2005); see also Nielsen et al. (2005).

The OECD report on environmentally sustainable transport (OECD, 2002: 96) advocates investment in rail infrastructure to serve both passenger and freight traffic: 'rail movement is inherently more efficient than movement by road, but its stronger feature could be the opportunity it provides for the use of renewable energy sources'. For a public transport system, sustainability depends on the total energy used to move people, as well, of course, as the energy used in the construction of vehicles, together with the source of that energy. Sustainable public transport systems require reasonably high levels of vehicle occupancy, and there can be a difficult trade-off between providing high levels of service throughout the day which can attract users out of their cars, and high levels of vehicle occupancy. But, as the OECD report points out, this does not matter if all the energy used to run the system comes from renewable sources. This is so much easier to accomplish with trains and even buses than with cars. There are also perfectly viable methods of providing efficient public transport services to dispersed – low density – urban environments as described in Chapter 9 of this volume and by Mees (2000, 2010).

A further important characteristic of public transport is that it is much more compatible with a high quality of urban place. For instance, in many European cities, trams, with their predictable – if speed restricted – movements, can mix with pedestrians in walkable spaces – think of the Bahnhofstrasse in Zürich, Switzerland or the Place de la Comédie in Montpellier, France, shown respectively by images 4.1.7 and 4.2.5 on the Companion Website. Even where, for various reasons, there needs to be some separation between trams or buses and pedestrians, the public transport modes use up much less of the public space per person transported than do cars. Nevertheless enthusiasm for public transport needs to be qualified. Maybe, as Hamilton remarks:

> Public transport is not the answer. It will of course have its part to play, but if we are to move on from the individuated, socially divisive and environmentally damaging way of life that accompanies car dependence, then we are simply going to have to travel less.
>
> Hamilton, 2003: 59

And we had better make sure that good public transport opportunities are provided across whole cities and do not become the preserve of the newly affluent, denser inner cores (Schaeffer and Sclar, 2003).

Walking

(see Images 4.2.1 to 4.2.10 in the Companion Website)
Walking is a mode of transport in its own right, not just an ancillary to motorized transport. Tolley's edited book (Tolley, 2003) gathers the work of many writers on sustainability and active transport. The place of walking in sustainable transport is clearly enunciated for instance by

Rees (2003), Hillman (2003), Monheim (2003), and Gehl and Gemzoe (2003) from the varying perspectives of ecology, public policy, transport planning and urban design. They are joined in the volume by many other insightful writers and practitioners. Walking as a transport mode is particularly important in poorer countries where the majority of trips are made on foot, and car travel is a distant aspiration (Simon, 1996; Imran, 2010). As Simon (1996: 95) remarks, ‘The vital importance of walking, especially over comparatively short distances, in cities of all sizes is often overlooked. Indeed many people do not even consider it to be a form of transport’. The result is that walking is often mixed dangerously with motorized transport, and walkers are left with no safe space. The same applies even more to people who get around by self-powered wheelchair. Even in rich cities, walkers often vastly outnumber motorists on busy streets but are provided with a much smaller space in which to move. Mees (2010: 184) points out that the neglect of walking as a transport mode leads to underestimation of its importance. He notes that in the journey to work in Zürich, ‘virtually all the public transport users walk to the tram, train or bus stop’ (ibid.). So the share of walking for at least some part of the journey is ‘more like 76%’ rather than the 10% recorded as the “walk to work” mode’. So improving the walking environment can encourage public transport use.

Walking is self-evidently the most sustainable mode of transport. Walking, along with sitting, standing and strolling is a social activity. It is essential to the enjoyment of public space and part of every trip, motorized or otherwise. Sustainable transport means reversing the order of priorities: first private vehicle transport, second public transport, and third active transport – with walking at the bottom of the heap (Low et al. 2005: 137). The first priority in cities is to make sure that walking is a safe and enjoyable experience – and to invest public money in infrastructure (streets and squares) to make it so. Kitahara (2003), writing about the small city of Kyojima in Japan, describes the particular qualities that create a walkable neighbourhood: low rise and openness of houses, moderately narrow streets and alleys, places where people come together and socialize. He says continual community engagement and pressure is necessary to protect such neighbourhoods: ‘It is necessary to gradually renew a community organization through educating and involving the younger residents in the project’ (ibid.: 509). Fenton (2003) also argues that individual interventions to encourage people to walk are not sufficient to stop people reverting to the habit of car use. Rather, collective activities and government interventions to build infrastructure for safe and attractive space, to increase public knowledge and improve public transport are essential to maintain walking for public health. We argue in Chapter 8 that alliances of communities and governments are necessary to transform transport towards sustainability.

Cycling

(see Images 4.3.1 to 4.3.24 and documents 4.3.1 and 4.3.2 in the Companion Website)

As with increased walking, measures to increase cycling for routine trips are part of most programmes for sustainable transport (McClintock, 2002; Tolley, 2003; Horton et al., 2007). Such measures include providing infrastructure specifically designed for cycling – whenever possible separated from motorized traffic – making sure that the cycle path system is a fully connected network without breaks, reducing the speed of motorized vehicles on roads shared by cyclists (PACTS, 1996), providing storage at destinations for bicycles, and combining cycling infrastructure with the public transport network. Some argue that cycling safety is best achieved by having many more cyclists sharing the road with cars. But that will not happen if people feel that sharing the road is unsafe: Catch 22! A plausible indicator of safe cycling environments is the number of women who use them. In the use of vehicles everybody makes mistakes, so as Welleman (2002: 203) argues, 'A truly safe infrastructure will almost automatically lead to the desired form of behaviour of traffic participants'.

The bicycle is possibly the most revolutionary form of transport ever invented. For short distances – 5–7 kilometres – in cities with even minimum congestion it rivals the average speed of cars. It extends the range possible for an average person within 30 minutes from 2.5 to up to 15 kilometres. It can carry surprising amounts of goods. And it does this without a large capital outlay, with almost zero carbon emissions, and with significant benefits to human health – from regular exercise. The cost of providing safe infrastructure for cycling is a small fraction of the amount most countries spend on roads for motor vehicles. With these advantages and a world on the brink of dangerous climate change it is surely negligent not to provide such infrastructure (see Documents 4.3.1 and 4.3.2 for an Australian analysis).

Mees (2010: 189) cautions that an increase in the proportion of journeys made by bicycle within a city or region does not necessarily indicate a step towards sustainability. The question, he writes, should be: 'was this at the expense of the car?' The question is a good one if the proportional use of different transport modes is used to assess sustainability – as it sometimes is. However, the best indicator of sustainability against the Principles is the overall performance of the transport system in terms of GHG emissions. If people choose to cycle rather than take public transport – for instance as seems to be happening in Paris with the Velib system – this will free up space on buses and trains – in Paris the Metro – adding little or nothing to the system's emission load, even though it may not be reducing it. Where public transport systems are overcrowded, with crammed trains, trams and buses, reduced crowding could encourage more people to use them. Also, as Mees acknowledges, cycle networks linked to public transport nodes can assist public transport use in low-density areas, substituting bicycles for cars. This requires good storage facilities at stations for bikes, normally lockable and under cover. Additionally, making places safer for cycling for both recreational and utilitarian purposes is an important part of the broader vision of quality of place associated with sustainable transport, as we discuss below.

Sustainable automobility

The prospect of automobility by other means is extremely attractive to existing business interests and governments because all that seems to need to change is technology. The vehicle and roads industries can remain exactly as they are. The technology for electric cars has been around for at least 120 years (Flink, 1988: 8). Hybrids, hydrogen fuel cells, and alternative – renewable – fuels have been much discussed, and it seems probable that sustainable automobility will eventually play some part in sustainable transport. The potential of such a position has already been discussed in Chapter 3 and little need be added here.

The jury is still out on whether the electric car can be a substitute for the world's internal combustion engine car fleet. At present it seems unlikely that electric cars can become a *sustainable* mass transport mode. The problem is not the electric drive but the storage on the vehicle: the battery, and the energy source of the electricity. Black (2010: 238) considers the possibility that recharging could be done at low demand hours, but acknowledges the uncertainty about whether this could be done without requiring additional power production – and hence increased emissions. We do not yet know whether batteries for electric cars can be produced sustainably or in sufficient numbers to allow substitution on the vast scale of the world's car fleet. Much depends on what they are made of, the availability of the supply of the necessary minerals batteries need, and the emissions cost of resource extraction, manufacture and disposal. There are also major institutional problems. Although some electric vehicle promoters insist that the ultimate source of electricity must be renewable energy, governments will have to legislate to make that compulsory. If left to the market, the cheapest alternative will prevail, and without regulation or incentives that put a high price on carbon emissions, that will remain fossil fuel for many years.

With alternative fuels, for sustainability what one would have to look for is a biological fuel source that does not compete with food production in the use of land. Such a possibility of mass fuel production from such a source perhaps exists in the future – for example oil from algae. How distant a future remains unknown.

Low carbon logistics

Compared with sustainable passenger transport, sustainable freight and logistics occupies a much smaller place in the sustainable transport literature. Nevertheless, as will have been apparent from the discussion in

Chapters 1 and 2, the problems of the impact of freight transport are global in scope, severe and difficult to solve. Whitelegg (1997: 21) noted that, 'Freight transport issues are central to the achievement of sustainable transport', and he devotes a chapter to them as well as 'solutions' (ibid.: 211–19). The solutions he advocates involve reducing freight travel by the substitution of 'near suppliers' for 'far suppliers' and the elimination of 'useless transport' – 'circuitous journeys dictated by the pattern of distribution centres and national/international supply chains'. Whitelegg cites research by Strutynski (1995), Holzapfel (1995) and Böge (1995) that estimates that a 50 per cent reduction in distance travelled by goods could be achieved just by substituting 'near' for 'far' suppliers. Further savings in emissions – in the UK – can be achieved by a staged modal shift:

i utilizing rail/road intermodalism and transferring 50–90 per cent of motorway freight onto the railways – cities with tram systems might consider the use of trams as freight carriers (Rein and Roggenkamp, 1995, with an example from Kassel, Germany);
ii utilizing port and waterway capacity to achieve higher levels of transfer away from road freight;
iii reconstructing logistics and advancing the idea of regional production systems to minimize distance travelled by goods (Whitelegg, 1997: 212).

Unfortunately these practices come up against the neo-liberal model of 'least cost' global economics in which mobility is used to overcome distance, production is sited wherever cheap and appropriately skilled labour is available, and environmental cost is discounted. Strutynski (1995) concedes that a five-fold increase in the cost of road freight transport would be necessary to motivate the shift.

Schiller et al. (2010: Chapter 5) provide an excellent update on sustainable logistics and freight. Though they don't cite Whitelegg's pathbreaking work, essentially the prescriptions are similar though some new ones are added:

- Increasing the energy efficiency of freight vehicles.
- Differentiation of the fleet – using smaller vehicles with smaller engines (three wheelers) or even foot-powered cargo bicycles for final delivery.
- Improved intermodal services and facilities.
- Shifting the truck–train breakeven point in favour of rail transport.
- Reforming ocean shipping to reduce fuel consumption.
- Shifting some airfreight to high-speed rail.
- Making companies responsible for recycling the packaging of the goods they sell.
- Reducing unnecessary freight travel and shortening supply chains through fiscal incentives. (Schiller et al., 2010: 140)

Schiller et al. (2010: 150) recognize that, 'Globalization, with its heavy dependence upon long-distance supply chains, works against sustainable development'. Global business as usual entrenches what Whitelegg has called a 'distance intensive economy'. No one has really explored the social and economic consequences of a fundamental change in the global business model. Yet, if climate change is to be mitigated at all, that model may have to change.

The above set of practices itself goes some way to creating a 'vision' of sustainable transport. But something is missing. If we return to the 'cyclone' of automobility, what sets it in motion, and what still partly underpins it is human desire: for the freedom, flexibility, excitement and delight of the self-contained vehicle on the open road. That is a vision. At the start of automobility, cars were expensive and exclusive toys to expand the lifestyle of the rich. Now almost everyone has a car and, despite the best endeavours of the advertising industry, the real experience of mass ownership and use of cars is often daily drudgery. Yes, rich people can own more expensive and luxurious cars but these are only marginally different from the average and their drivers get stuck in traffic like everyone else. Gadgets and leather seat covers are no substitute for the novelty of speed on the open road. The key here, though, is 'lifestyle' which generates aspiration. Aspiration, as we saw, led within 50 years to mass consumption entrenching a vast range of industries and types of employment. Does 'sustainable transport' contain such a vision that could create aspiration in such a way as to support the public and private investment necessary to realize it?

The outcome of using cars for daily travel was apparent at least 40 years ago. In the 1970s Bendixson (1974: 12) wrote, 'The problems are everywhere the same: the incompatibility of giant highways with decent living conditions, and the liquidation of public transport by an advancing army of cars'. Opinions on the car have changed, but, he continues, 'Government response to this changing opinion has in most cases been slow. Many city engineers still believe there is no alternative to catering for increased car use, and the same view remains well entrenched in all ministries of transport' (ibid.). What happened to cities is that quality of place was subordinated to automobility, especially in the suburbs where most people live. The vision of sustainable transport cannot offer a complete parallel to the vision of automobility, based on a single artifact: the automobile. But it does offer to create or restore something of great value. That can be expressed as 'quality of place' – less tangible than the quality of a thing – the best that city living has to offer. Of course it is a subjective concept, but then 'visions' have to be subjective, appealing at once to emotions and ideas to inspire action. While automobility offered fun and escape from the city to the imagined delights of the countryside, sustainable transport offers to make city life itself delightful. Quality of place is not reducible to sustainability or vice versa, but it is more than just a co-benefit. It is a key part of what the vision of sustainable transport offers to the public.

Of course, there are many possible variants of 'quality of place' limited only by subjective parameters which can probably only be found by practical trial and testing. They should be ultimately determined by the residents and workers of the place in question, which introduces the importance of participation in knowledge formation (discussed in Chapter 11). But still one can imagine an environment whose buildings are mostly of human scale and where the natural world of trees, vegetation and water is allowed to flourish, where the sort of things that people like to do in public space are catered for: shopping, sitting around having a coffee or a beer, and where places of work are not too far from homes. The transport dimension of this kind of place requires a network of paths so that movement by foot power is a pleasant and safe possibility, connecting lower density areas with activity nodes and terminals of the rapid transport network that connects the place to the rest of the wider city by train, light rail (trams), and buses. This is the sort of place where many people seem to want to live today; a place where public space is valued and just as attractive as the space in private homes and gardens. This is a place where people can move between private and public space and feel a sense of collective ownership and pride.

This vision of place was developed by those seeking to reduce 'automobile dependence'. It became the 'new urbanism' which Newman and Kenworthy – themselves early advocates of quality of place – describe as:

> A movement that incorporates the need to expose car-dependent assumptions in town planning rules and fashions; it orients instead around a transit system and attempts to create walking environments through denser, more mixed land use, houses fronting streets with garages behind and other design qualities.
>
> Newman and Kenworthy, 1999: 143

Unfortunately the immensely important vision of quality of place has become clouded by debates about urban density, with the unfortunate implication that sustainable transport requires the wholesale reshaping of cities. We agree with Mees (2010: 53):

> If densities of 400, 240 or even 100 [persons] per hectare are really required to make transport sustainable, then we had better begin searching for a new planet to live on. Built form changes very slowly in aggregate – most of the houses New Yorkers or Melbournians will inhabit in 2050 already exist – but climate change requires urgent action, and the other problems of automobile dependence cannot be put off for decades either.

There are, in fact, mutually indispensible visions: quality of place and sustainable transport, but they do not reduce to one another. They respond to different criteria. Measures of density do not provide the bridge between them. Quality of place is the positive opportunity that is an essential counterpart to the negative risk of global warming and the rising oil price. The vision must offer not only escape from the negatives but also something positive and desired. But still it has to be connected back to the principles of sustainable transport stated above. To do that requires measurement of the carbon impact of urban mobility and the vehicles and infrastructures to support that mobility (Hickman et al., 2010). Cities, in the developed world at least, already have embedded, path dependent, spatial forms and are already equipped with transport systems and infrastructure. Change will occur only gradually, step by step, and new transport systems aligned more strongly to the principles of sustainability will need to evolve out of the old systems already in place. Every city is different in this respect. Every city population has different ideas about quality of place. Questions of quality of place and sustainable transport need to be opened up to the public, because it is the public and not the academy that will make change occur.

Both Newman and Kenworthy (1999) and Whitelegg (1997) point out that there is nothing inevitable about automobility (auto-dependence) and unsustainable transport. Newman and Kenworthy (1999: 129) describe ten myths of the inevitability of automobile dependence. We completely agree with these authors. We capture some of the more compelling 'myths' as self-perpetuating storylines about automobility (Curtis and Low, 2012). It may seem in the chapters to follow that by giving attention to the barriers to sustainable transport we are merely further indulging the myth of inevitability. This is not the case, as we hope to show. Our intention is rather to elucidate the barriers in order to broaden the range of political strategies needed to overcome them. That said, as was apparent in the first section of this book, the dilemma is a formidable one of global dimensions, and if we fail to pay attention to it, and to the political strategies required to resolve it, we may be left with nothing but sustainability as political compromise – with some excellent but very patchy local results, without the global transformation in time to avoid dangerous climate change. The evidence of 'ecosocialisation' is growing in which the world's economic and social behaviour is gradually being adapted to conservation of the biosphere under pressure from the triple social movements for global markets, social protection and ecological conservation (Low, 2002). But is that process moving fast enough? Those who care about sustainable transport should not feel comfortable with the speed of progress.

Sustainable transport in market economies

While Principle 1 above can perhaps represent 'environmental sustainability' in conventional thinking, and Principle 2 'social sustainability', 'economic sustainability' has not only been demoted but has been deliberately omitted. This does not mean that there should be no engagement with businesses with an interest in developing the 'green economy'. On the contrary such engagement is necessary. But far too often 'economic sustainability' means sustaining 'business as usual', resulting in a weak compromise, rather than transforming the economy. Of course economic growth and prosperity of national economies within a global regulatory framework are important goals. But they will automatically be sought by governments of market economies. Sustaining the economy is structured into liberal or social democracies because governments and businesses are locked together in a common interest in economic growth – governments because they are dependent on votes, as voters are dependent on employment provided by companies. Even where there is no electoral democracy – as in China – governments are still dependent on the consent of the people.

Charles Lindblom (1977), the doyen of American pluralist political theorists and co-author with Robert Dahl of the concept of polyarchy, argued persuasively that business – or capital in Marx's terms – occupies a privileged position with respect to government:

> Any government official who understands the requirements of his position and the responsibilities that market-oriented systems throw on businessmen will therefore grant them a privileged position. He does not have to be bribed, duped, or pressured to do so. Nor does he have to be an uncritical admirer of businessmen to do so. He simply understands, as is plain to see, that public affairs in market-oriented systems are in the hands of two groups of leaders, government and business, who must collaborate, and that to make the system work government leadership must often defer to business leadership. Collaboration and deference between the two are at the heart of politics in such systems.
>
> Lindblom, 1977: 175

The problem, Lindblom argues, is that business can in effect veto not only redistributive measures but also 'proposed solutions to collective problems' (Lindblom, 1977: 347). Importantly, the privileged position of business 'permits it to obstruct policies such as those on environmental pollution and decay, energy shortage, inflation and unemployment, and distribution of wealth' (ibid.).

Since 2008, as governments have hurried to spend billions of taxpayers' money propping up banks and the motor industry, Lindblom's analysis has had particular and renewed resonance. Economic growth and sustainable transport are two essentially different goals, and they should be kept conceptually separate. Inevitably the campaign for sustainable transport will be required to show how its practices may be made compatible with economic growth (see for example Jacobs, 1991 and Eckersley, 2004 for useful theoretical discussions on this topic), but that is quite different from building economic growth into sustainable transport as an a priori principle, let alone as the dominant principle.

Technical efficiency is another matter again. In pursuing the principles and programme of sustainable transport it is obviously sensible to adopt practices that can meet the principles at least cost. But that is an instrumental objective subordinate to the principles. It may not be possible to meet the principles in time to be effective. Nazism would not have been defeated if the production of aircraft, ships and tanks had been left to price incentives set by government fiscal policy. Carbon trading may in theory deliver least cost outcomes, but it is also more complex to administer and more open to corruption than a simple tax. Technical efficiency can all too easily displace the principles along the lines of this false syllogism: 'market solutions are the best way to guarantee that least cost solutions are pursued, therefore market solutions are the ones that should always be adopted'.

Automobility was never simply a market-based solution. In fact, the vision was implemented with a great deal of help from government intervention – or 'Fordism' as we discuss in the next chapter. But the vision of automobility is strongly compatible with the disaggregated ideology of market capitalism: individuals finding their own solutions in pursuit of a goal – mobility – organized only by price, within a broad regulatory framework provided by governments. At a micro level, government interventions – for example, providing roads and safety regulations – could be justified by the need to facilitate individual mobility solutions. Governments could look at trends expressed in actual travel behaviour and simply provide for their continuation. Congestion was treated as evidence of demand.

Sustainable transport at its core is very different from automobility in relation to market capitalist societies – whether democratic or otherwise. Sustainable transport, with placemaking and urbanity as its core vision, requires public planning, a concept that has lost favour with the advance of neo-liberalism. The very terms 'public' or 'collective' transport clash with the individualism at the heart of the neo-liberal project. It is no coincidence that many of the paradigmatic examples of placemaking and sustainable transport are to be found in the social democracies of Europe and South America. So, instituting sustainable transport as the dominant mobility paradigm faces considerable barriers. In Chapter 5, we analyse some of the complexities of global regimes of regulation of market capitalism – Fordism and neo-liberalism. In Chapter 6, we examine at a theoretical level, some other institutional barriers to sustainable transport.

Conclusion

In this chapter we have examined a process, discussed sustainability as a political compromise, a prudential economic principle and a moral imperative. Two clear principles of sustainable transport have been posited. Under the rubric of the latter we have summarized some of the practices necessary to implement the principle, and postulated that their achievement will require concerted struggles to overcome barriers. These barriers are formidable. The array of industrial and labour interests in maintaining the status quo is vast. The task of industrial restructuring for sustainable transport is therefore enormous, but has hardly been touched on in the literature to date (but see Weaver, 1998). The introduction of a carbon price, either through a tax or an emissions trading scheme is unlikely by itself to have much impact on travel behaviour or choice of transport mode. The price of fuel will have a larger impact (see Document 4.1 in the Companion Website for an Australian analysis). In response to price increases attractive alternative transport options must be available in order for people's travel behaviour to change.

The process through which automobility became established as the dominant paradigm for mobility cannot be simply replicated for sustainable transport. But understanding processes – socio-technical processes that are deeply political involving social and economic change – can help us to understand what is involved in transformation from one paradigm of mobility to another. Automobility did not occur without aspiration and popular struggle, and those elements will almost certainly have to be present in the transformation to sustainable transport. A broad coalition of interests will have to be constructed around both the positive goals – achieving a better quality of life in urban space – as well as the negative – avoiding threats such as climate change and reduced mobility resulting from escalating fuel prices.

Unlike the struggle for automobility, the struggle for sustainable transport will not find an easy accommodation with the dominant politico-economic ideology. On the contrary, moving to sustainable transport requires a kind of thoughtful, consultative and intelligent public planning which is the antithesis of the 'leave it to the market' doctrine of neo-liberalism. In some places that kind of planning will have to be revived or reinvented as a precondition for sustainable transport. Nevertheless at the time of writing the world seems to be on the cusp of change. The 'global financial crisis' starting in 2008 is only the latest in a series of escalating economic crises, each more severe than the one before. If history is any guide to the future, the incapacity of the dominant ideology to resolve the fissures in capitalism will lead to a moment favourable to transformation. In the next chapter we develop a framework within which to explore such critical junctures for transformation in an East Asian context, that of Japan.

Notes

1 The term 'epistemic community' refers to 'a network of professionals with recognized expertise and competence in a particular domain and an authoritative claim to policy-relevant knowledge within that domain or issue area (Haas, 1992: 3).

2 We hasten to add that Professor Black is not one of them.